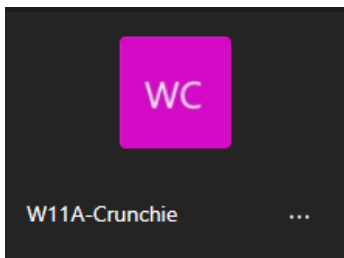


W11A-Crunchie

Requirements Elicitation, Analysis & Specification

Report



Presented by

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[Requirements] Elicitation:

Target User Interviews:

Name:	Email:
Dean Fellner	Dean.fellner@gmail.com
Joshua Gessner	Joshua.gessner5@gmail.com
Sadman Hyder	thesadmanhyder@gmail.com

Interview Questions and Responses:

1. **What challenges do you face with your current quiz tools that you hoped Toohak would solve?**

Dean Fellener's Response:

A: The challenges I face with current quiz tools include a lack of customisable and interactive features in the quizzes. I was hoping Toohak would offer more features to make the quizzes more engaging compared to traditional quiz platforms.

Joshua Gessner's Response:

A: Challenges I face mainly include the inability to host huge lobbies, the inability to select my own background music and audio, the inability to have sound-based questions such as "What animal is this", playing animal calls and can't make team games

Sadman Hyder's Response:

A: The current list of customisation options for the app is quite limited as only includes being able to insert the question and a very short description. There aren't any options to change the colour, text size or font of the questions.

2. **Are there any features you wish Toohak had to better address your quiz creation and management needs?**

Dean Feller's Response:

A: I wish Toohak had improved its UI and UX capabilities. Current quiz tools on the market are very user-friendly, but Toohak isn't quite at that level yet, though I understand it's still early in development. I also wish it offered more customisable and interactive features (as mentioned in Q1) to make quizzes more engaging for users.

Joshua Gessner's Response:

A: Features I would like to see are the ability to host super lobbies, create teams, add my own background music, have sound-based questions, have video-based questions and have lighting rounds where you have 5 seconds to pick an answer with more intense music.

Sadman Hyder's Response:

A: Could be better if the quiz-creating process were more flexible (e.g. does not require a thumbnail). It is also a bit ambiguous. Are the answers multiple-choice? Or is it a short answer question where the correct answer is notated in the tick and the wrong answers in the cross? What about answers that are not included? Are they assumed to be incorrect?

3. **Were there any features within Toohak that you thought were well-integrated for creating and managing quizzes?**

Dean Feller's Response:

A: The navigation bar at the bottom was well-integrated, making it very simple and easy to create a new quiz. The design for creating quizzes was extremely simple, which made it easy to use. However, it felt basic and lacked the UI/UX polish compared to other competitors.

Joshua Gessner's Response:

A: The ability to transfer quizzes effectively is a nice touch, also the ability to restore deleted quizzes is very nice, so for me overall, the features are well-integrated enough for me to actually create quizzes and play a game with friends for my own leisure.

Sadman Hyder's Response:

A: The best feature is that an unfinished quiz is automatically saved for you so you do not randomly lose your progress. Also, the trash bin is also really useful in case you want to recover and the different clickable options are easy to navigate through for the most part. Creating new questions and editing and deleting them is relatively simple.

4. Did you find the user interface of the Tahook quiz game intuitive and easy to use? Why and why not?

Dean Feller's Response:

A: Yes, the user interface was simple and easy to use. However, I encountered a few issues, such as not being allowed to write answers and questions longer than 30/50 characters. On the quiz list, it would be helpful if the emojis (trash and send) displayed what they are when hovering over them. Also, having an edit button on the homepage for the quizzes would be useful.

Joshua Gessner's Response:

A: It's ok, it's not the best in the world but not the worst. It could very obviously do some work in the looks department but in an overall sense, it's functional, serves the intended purpose and since the screen isn't cluttered, it was actually very easy to use.

Sadman Hyder's Response:

A: It is intuitive to use. Pretty easy to navigate considering it only has four tabs. Can't really get lost in there. However, the interface can be made to look more appealing, as of right now, the interface gives off a very unprofessional vibe to me, to be honest.

5. Would you recommend the Toohak quiz game to others? Why or why not?

Dean Feller's Response:

A: I would not recommend the Toohak quiz in its current stage of development, mainly due to the UI and UX not being as polished as competitors like Quizlet and Kahoot. However, if Toohak improves its UI and UX, I might consider recommending it because of its simplicity and ease of use.

Joshua Gessner's Response:

A: Yes even though it's in a rough stage right now, the project looks like it will be a useful tool in the future with lots of interesting features to come, so I am optimistic about this being a success, provided that the right updates are implemented to make the game an even better alternative to other quiz tools such as Kahoot.

Sadman Hyder's Response:

A: I would recommend it if there are some improvements made to the interface design and there are more customisation options provided to make my experience more fun and memorable. So all in all, in my honest opinion, this would be excellent for people who want a quick and easy-to-use quiz creation tool.

Problem Identification and Solving:

Problem 1: Lack of Customisable and Interactive Features

Description of the Problem: Toohak's current quiz tools are limited in terms of customisation and interactivity, which restricts user engagement and the variety of quiz experiences for the player and other guest players.

- **Proposed Solution:** Introducing more customisation options and interactive features in Toohak, such as the ability to select and upload their own background music, multimedia support in the form of audio and video features, text customisation for size, font and colour and integrating interactive elements like timers, animations & team-games would help improve the overall user experience.

Problem 2: Hosting Limitations for Large Lobbies

Description of the Problem: Players in Toohak are currently unable to host large quiz lobbies, limiting the number of players that can be added to a single quiz game session. This would affect individuals such as teachers or party hosts who would use such features to host a large-scale quiz game to engage with a wider audience.

- **Proposed Solution:** Toohak's current backend infrastructure can be further optimised to support large-scale quiz session lobbies in the form of implementing scalable server architecture to handle the increase in traffic and adding a feature that can help the host with keeping track of players and their responses in a larger game lobby.

Problem 3: Ambiguous Quiz Creation Process

Description of the Problem: The current quiz creation process lacks clarity on whether answers are multiple-choice or short answers, which makes the creation of questions confusing. Additionally, the requirement for thumbnails may be unnecessary for some users, as a default thumbnail option might save time rather than uploading a thumbnail from the drive.

- **Proposed Solution:** Toohak can provide clearer options for question variety so that players can specify what question selection types to pick when creating a quiz. Moreover, including a step-by-step tutorial for new players explaining how the quiz creation process works would help alleviate any further confusion. Lastly, thumbnails should be made optional for users to decide from.

Problem 4: Basic and Unpolished UX/UX and Navigation

Description of the Problem: The current user interface for Tahook is commented to be unprofessional in comparison to its competitors, with issues such as character limits for questions and answers, unclear icon functions, and a lack of essential editing features limiting intuition.

- **Proposed Solution:** Redesigning the current UI/UX to improve functionality and to make it more user-friendly would allow for a more approachable experience for players in navigating Tahook's quiz tools. So as such, this would involve expanding character limits, enhancing current icons to be more vibrant and implementing an edit button on the quiz's homepage to refine the overall visual appeal of Toohak.

Problem 5: Lack of Advanced Quiz Management Features

Description of the Problem: Despite the appreciation for having useful quiz management features such as quiz transfer and restoration, users commented on wishing that there was better handling of unfinished quizzes, clearer options for quiz recovery, and improved overall quiz management.

- **Proposed Solution:** When players are creating a quiz, an auto-save feature that periodically saves quiz progress and shows up as a text-box notification on the side would ensure players that their quizzes are being saved. Also, providing a centralised dashboard where users can see the status of all their quizzes (e.g., drafts, published, trashed) would greatly help streamline quiz organisation and management.

User Stories:

User Story 1 Name: Customisable and Interactive Features

Priority	HIGH
Estimate	9 days
User Story	As Hriday (a player), I want to enhance my quiz's atmosphere with a 1980s pop theme to the questions, timer and animations while also having Michael Jackson's Thriller song playing in the background to liven up the mood for my guests.
Acceptance Criteria	● Players can upload and select background music from their personal library or from the internet. ● Players can add and play audio/video files within questions to their liking. ● Players can configure and display timers and animations to their chosen theme. ● The features are accessible and user-friendly for players to utilise at their own leisure.

User Story 2 Name: Larger Lobby Hosting

Priority	HIGH
Estimate	7 days
User Story	As Hayden (a teacher/host), I want to host a large quiz lobby to accommodate a lecture full of university students so that they can all join in together for a single fun and educational quiz session.
Acceptance Criteria	• Tohaak's system can support up to 300 students at max capacity. • Hosts are able to view and manage the list of participants active within their respective quiz sessions. • System performance remains stable with large lobbies. • Hosts can track participant responses effectively.

User Story 3 Name: Clear Quiz Creation Process

Priority	MEDIUM
Estimate	4 days
User Story	As Daniel (a player), I want an online guide to show me how to customise my questions and decide between picking my thumbnail or default to simplify the quiz creation process for myself.
Acceptance Criteria	• The question types (multiple-choice, short answer) are specified and selectable for the player to choose from. • Thumbnails are optional with default options available for the player.

	• A step-by-step tutorial guides new users through the quiz creation process.
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User Story 4 Name: Enhanced UI/UX and Navigation

Priority	HIGH
Estimate	6 days
User Story	As Itquan (a player), I want an improved user interface with easy navigation and modern aesthetics to make Tohaak more user-friendly and visually appealing.
Acceptance Criteria	• Tahook's user interface is redesigned with expanded character limits for questions and answers. • An edit button is available on the homepage for each quiz. • Icons are vibrant and intuitive for the player to navigate with • Navigation enhancements improve overall usability.

User Story 5 Name: Advanced Quiz Management

Priority	HIGH
Estimate	6 days
User Story	As Eakveer (a player), I want to see the status of all the quizzes I have made so far in Toohak so that I can finish the incomplete or deleted ones and then decide if I want to update their progress or not before going to

	sleep.
Acceptance Criteria	• Auto-save feature periodically saves quiz progress. • Notifications inform users when progress is auto-saved. • A centralised dashboard shows the status of all quizzes (e.g., drafts, published, trashed). • Enhanced options for recovering unfinished or trashed quizzes

[Requirements] Analysis & Specification - Use Cases

Use Case 1: Customise Quiz with 1980s Pop Theme

Priority	High
Actors	Primary Actor: The player Secondary Actor: Toohak Quiz System
Trigger	The player logs into the Toohak system and starts customising a Toohak quiz.
Preconditions	• The player is logged into the Toohak system. • The player has an existing quiz or is creating a new quiz centered around Pop knowledge. • The player has access to a Michael Jackson Thriller music file. • The system is online and accessible.

Basic Flow

Step	Actor Action	System Response
1	The player logs into the Toohak system.	The system validates the player's username/password and grants them access.
2	The player creates a new quiz.	The system displays the quiz customisation options.
3	The player selects a theme that is similar to 1980s pop music from the available themes.	The system applies the theme to the quiz, updating the visuals and animations accordingly.
4	The player uploads Michael Jackson's Thriller song from his personal library.	The system validates the upload process and sets the uploaded song as the background music for the quiz.
5	The player selects a question and chooses to add an audio or video file.	The player uploads the desired audio/video file so that the system can validate it and link the file with the chosen question.
6	The player configures the timer settings to match the 1980s pop theme.	The system updates the timer and animations for the quiz.
7	The player previews the customised quiz.	The system displays the quiz with all customisations applied, including the background music, themed timer, and animations.
8	The player saves the customised	The system confirms

	quiz in their quiz library.	the save and updates the quiz in the database.
Alternative Flows		
	Step	Condition
	4a	Invalid File Format
		1. The system returns an error message and prompts the player to upload a valid file. 2. The player uploads a valid file.
	4b	The file is too large
		1. The system returns an error message and prompts the player to upload a smaller file. 2. The player uploads a smaller file.
	5a	Network error during file upload
		1. The system returns an error message and prompts the player to retry. 2. The player retries the file upload.
	5b	The player chooses to add an audio/video file from the internet
		1. The player selects a file from the internet. 2. The system validates and processes the file from the internet.
	7a	The player wants to make further changes after previewing the quiz
		1. The player makes additional customisations.

	<table><tr><td></td><td></td><td>2. The system updates the quiz with the new customisations.</td></tr></table>			2. The system updates the quiz with the new customisations.									
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Network Error	There is a network error during the upload.	The system returns an error message and prompts the player to retry.											
Postconditions	• The player is logged into the Toohak system. • The player has an existing quiz or is creating a new quiz. • The player has access to the music file to upload. • The system is online and accessible.												

Use Case 2: Larger Lobby Hosting

Priority	High
Actors	Primary Actor: The Host

	Secondary Actor: Toohak Quiz System		
Trigger	The host logs into the Toohak system and starts hosting a quiz session.		
Preconditions	• The host is logged into the Toohak system. • The system is online and accessible.		
Basic Flow			
	Step	Actor Action	System Response
	1	The player logs into the Toohak system	The system validates the player’s username/password and grants them access.
	2	The host starts a quiz session	The system displays options for managing the quiz lobby.
	3	The host configures the quiz to accommodate up to 300 participants.	The system updates the session settings to support a larger lobby.
	4	Students join the quiz session.	The system processes each join request and adds the student to the session.
	5	The host monitors and manages the list of participants.	The system displays the list of active participants and their statuses.
	6	The host tracks participants' responses during the quiz session.	The system collects and displays responses in real time.

Alternative Flows	<table><tr><th>Step</th><th>Condition</th><th>Alternative Flow</th></tr><tr><td>3a</td><td>The system cannot support the desired number of participants due to limitations.</td><td> 1. The system returns an error message and prompts the host to reduce the number of participants. 2. The host adjusts the participant limit. </td></tr></table>	Step	Condition	Alternative Flow	3a	The system cannot support the desired number of participants due to limitations.	1. The system returns an error message and prompts the host to reduce the number of participants. 2. The host adjusts the participant limit.			
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Invalid Participant Data	A participant provides invalid data when joining.	The system returns an error message and prompts the participant to correct their data.								
Postconditions	• The quiz lobby can accommodate up to 300 students. • The host can manage and view participants. • The system performance remains stable. • The host can track participant responses effectively.									

Use Case 3: Clear Quiz Creation Process

Priority	Medium																						
Actors	Primary Actor: The player Secondary Actor: Toohak Quiz System																						
Trigger	The player starts the quiz creation process and seeks guidance on customising questions and choosing a thumbnail.																						
Preconditions	• The player is logged into the Toohak system. • The Toohak system is online and running.																						
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Postconditions	• The player is guided through the quiz creation process. • The player can choose between custom and default thumbnails. • The quiz creation process is simplified with clear instructions and options.			

Use Case 4: Enhanced UI/UX and Navigation

Priority	High		
Actors	Primary Actor: The player Secondary Actor: Toohak Quiz System		
Trigger	The player logs into Toohak and experiences the redesigned user interface with improved navigation.		
Preconditions	• The player is logged into the Toohak system. • The system is online and accessible.		
Basic Flow			
	Step	Actor Action	System Response
	1	The player logs into the Toohak system.	The system displays the redesigned homepage with modern aesthetics.
	2	The player views the quiz list on the homepage.	The system displays quizzes with expanded character limits for questions and answers.
	3	The player clicks the edit button next to a quiz.	The system opens the quiz editor with the

			ability to modify the quiz details
	4	The player navigates through different sections of the system using icons.	The system provides intuitive and vibrant icons for smooth navigation.
	5	The player uses the enhanced navigation features to find and interact with different parts of the system.	The system provides improved usability and user-friendly navigation options.
Alternative Flows			
	Step	Condition	Alternative Flow
	3a	The edit button is not visible on the homepage for the quizzes.	1. The system checks the visibility settings for the edit button. 2. The system displays the edit button or notifies the player of the issue.
	4a	The icons are not displaying correctly.	1. The system returns a default icon or error message. 2. The system attempts to reload the icons or prompts the player to report the issue.
	5a	Navigation features are not functioning as expected.	1. The system provides a fallback navigation option. 2. The system notifies the player of the issue and offers alternative navigation aids.

Exceptions	<table><tr><th>Exception</th><th>Condition</th><th>System Response</th></tr><tr><td>Missing Edit Button</td><td>The edit button is not present on the homepage.</td><td>The system checks and corrects the issue, or provides a message explaining the absence.</td></tr><tr><td>Icon Display Issues</td><td>Icons are not loading or are corrupted.</td><td>The system displays default icons and provides an option to report the problem.</td></tr><tr><td>Navigation Failure</td><td>Navigation features do not respond or are slow.</td><td>The system offers alternative navigation options and provides a message about the issue.</td></tr></table>			Exception	Condition	System Response	Missing Edit Button	The edit button is not present on the homepage.	The system checks and corrects the issue, or provides a message explaining the absence.	Icon Display Issues	Icons are not loading or are corrupted.	The system displays default icons and provides an option to report the problem.	Navigation Failure	Navigation features do not respond or are slow.	The system offers alternative navigation options and provides a message about the issue.
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Navigation Failure	Navigation features do not respond or are slow.	The system offers alternative navigation options and provides a message about the issue.													
Postconditions	• The player experiences a visually appealing and modernized user interface. • The player can easily navigate the system using vibrant and intuitive icons. • The player benefits from improved usability and functionality of navigation features.														

Use Case 5: Advanced Quiz Management

Priority	High
Actors	Primary Actor: The player Secondary Actor: Toohak Quiz System
Trigger	The player logs into Toohak to view and manage the status of his quizzes.

Preconditions	• The player is logged into the Toohak system. • The player has an existing quiz or is creating a new quiz centred around Pop knowledge. • The player has access to a Michael Jackson Thriller music file. • The system is online and accessible.																		
Basic Flow	<table><tr><th>Step</th><th>Actor Action</th><th>System Response</th></tr><tr><td>1</td><td>The player logs into the Toohak system.</td><td>The system displays the dashboard with quiz statuses.</td></tr><tr><td>2</td><td>The player navigates to the centralised dashboard.</td><td>The system shows the status of all quizzes, including drafts, published, and trashed.</td></tr><tr><td>3</td><td>The player views notifications for auto-saved progress.</td><td>The system displays notifications about recent auto-saves and progress updates.</td></tr><tr><td>4</td><td>The player selects a quiz from the dashboard to view or edit.</td><td>The system opens the selected quiz with options to update or recover the quiz.</td></tr><tr><td>5</td><td>The player uses the enhanced quiz management options to recover unfinished or trashed quizzes.</td><td>The system restores or updates the quiz based on user actions.</td></tr></table>	Step	Actor Action	System Response	1	The player logs into the Toohak system.	The system displays the dashboard with quiz statuses.	2	The player navigates to the centralised dashboard.	The system shows the status of all quizzes, including drafts, published, and trashed.	3	The player views notifications for auto-saved progress.	The system displays notifications about recent auto-saves and progress updates.	4	The player selects a quiz from the dashboard to view or edit.	The system opens the selected quiz with options to update or recover the quiz.	5	The player uses the enhanced quiz management options to recover unfinished or trashed quizzes.	The system restores or updates the quiz based on user actions.
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Alternative Flows

Step	Condition	Alternative Flow
2a	The dashboard does not load properly.	1. The system displays an error message. 2. The system attempts to reload the dashboard or offers alternative access.
3a	Notifications for auto-saved progress are missing.	1. The system checks for notification settings. 2. The system resends notifications or alerts the player manually.
4a	The selected quiz cannot be opened due to a system error.	1. The system displays an error message. 2. The system attempts to recover the quiz or notifies the player of the issue.
5a	Enhanced quiz management options for quiz recovery are not visible.	1. The system provides alternative methods to access recovery options. 2. The system notifies the player of the issue and suggests solutions.

Exceptions	<table><tr><th>Exception</th><th>Condition</th><th>System Response</th></tr><tr><td>Dashboard Load Error</td><td>The dashboard fails to load or shows incorrect data.</td><td>The system displays an error message and attempts to reload or correct the dashboard.</td></tr><tr><td>Missing Notifications</td><td>Auto-save notifications are not received.</td><td>The system checks the notification system and resends notifications or alerts the player.</td></tr><tr><td>Network Error</td><td>The player cannot access or recover a quiz.</td><td>The system displays an error message and attempts to resolve the issue or recover the quiz.</td></tr></table>	Exception	Condition	System Response	Dashboard Load Error	The dashboard fails to load or shows incorrect data.	The system displays an error message and attempts to reload or correct the dashboard.	Missing Notifications	Auto-save notifications are not received.	The system checks the notification system and resends notifications or alerts the player.	Network Error	The player cannot access or recover a quiz.	The system displays an error message and attempts to resolve the issue or recover the quiz.
	Exception	Condition	System Response										
	Dashboard Load Error	The dashboard fails to load or shows incorrect data.	The system displays an error message and attempts to reload or correct the dashboard.										
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Network Error	The player cannot access or recover a quiz.	The system displays an error message and attempts to resolve the issue or recover the quiz.											
Postconditions	• The player can view and manage the status of all his quizzes. • The system auto-saves quiz progress periodically and notifies the player of these saves. • The player can recover unfinished or trashed quizzes and update their status as needed.												

[Requirements] Validation:

User Feedback

Dean Fellner's Comments: Nowadays since such a quiz system excites people, more people would want to join a quiz session, and as such, a larger lobby could be useful and this would make it more competitive for people wanting to enjoy a good competition. So in my honest opinion, Use Case 2 would be what best describes the problem that I am trying to solve.

Joshua Gessner's Comment: These use cases are extremely good and go over all the problems I have stated in my original comment and provide all the services I set out to achieve by using this quiz tool. Moreover, Use Case 1 would be the case I would personally choose as it allows for custom soundtracks in the game of Toohak to be added, which was what I wanted to see being implemented in the first place as it's a really fun idea. However, I would expand use case 1 to also include quiz-specific audio so you can play sound identification games with friends or at school.

Sadman Hyder's Comment: Use Case 4 is the most appropriate as it would make navigation of the website much easier for users. This is one of the most important aspects of a website with regard to making it appealing for a user given how the basic flow seems straightforward and provides a seamless and easy experience when making quizzes. So for me overall, the post-conditions are what I wish to achieve by having a visually appealing user interface and easy navigation bars Music.

[Design] Interface Design:

Route	CRUD Method	Input Parameters	Return Object	Description of What it Does
/v1/leaderboard	GET	None	leaderboard data (array of player scores)	Retrieves the current leaderboard for all quizzes
/v1/player/{playerid}/feedback	POST	playerId, feedback (text, rating)	status (success/failure), feedback id	Submits feedback from a player

